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“AI-Enhanced IoT Enabled Smart Kitchen Safety and Temperature Controlled Fan Automation System”

A PROJECT REPORT

submitted by

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in partial fulfilment for the award of the degree of

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in

ELECTRICAL AND ELECTRONICS

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V.S.B. ENGINEERING COLLEGE, KARUR

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BONAFIDE CERTIFICATE

This is to certify that the project report titled “AI-Enhanced IoT Enabled Smart Kitchen Safety and Temperature Controlled Fan Automation System” is the bonafide work of **R BABITHA**(Register No: 922524105005), who carried out the project work under my supervision during the academic year 2025–2026, in partial fulfilment for the award of the degree of Bachelor of Engineering in Electrical and Electronics Engineering.

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ABSTRACT

AI-Enhanced IoT Enabled Smart Kitchen Safety and Temperature Controlled Fan Automation System is an intelligent kitchen automation system designed to improve safety, comfort, and energy efficiency. It continuously monitors the kitchen environment using sensors to detect temperature, gas leakage, smoke, and flame. Artificial Intelligence (AI) analyses the collected data to identify potential hazards and make smart decisions.

When the temperature exceeds a predefined limit, the system automatically turns on the exhaust or cooling fan. If gas leakage, smoke, or fire is detected, it instantly activates a buzzer, sends alert notifications to the user, and takes preventive actions to reduce risks. The IoT platform enables real-time remote monitoring and control through a mobile application built on the Blynk platform.

The system further incorporates AI-based occupancy detection to switch the fan OFF when the kitchen is empty, thereby improving energy efficiency, along with AI-based gas leak detection, smoke detection, and fire risk prediction that combine multiple sensor readings to provide early warnings. This system enhances kitchen safety, minimises human intervention, reduces energy consumption, and provides a reliable, scalable, and cost-effective smart home solution.

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Chapter 1 – Introduction

1.1 Background

The kitchen is one of the most hazard-prone areas of a home, with risks arising from open flames, LPG cylinders, overheating appliances, and unattended cooking. Conventional kitchens rely almost entirely on manual observation for safety, which is unreliable, especially when occupants are distracted or absent. Fires caused by gas leakage, overheating, and unattended flames remain a significant contributor to domestic accidents.

With the growth of the Internet of Things (IoT) and Artificial Intelligence (AI), it has become possible to build intelligent monitoring systems that continuously observe environmental parameters and respond automatically to unsafe conditions. The AI-Enhanced IoT Enabled Smart Kitchen Safety and Temperature Controlled Fan Automation System is designed to address this need by combining an ESP32 microcontroller with temperature, gas, and flame sensors to create a self-monitoring, self-acting kitchen safety solution.

The system uses a DHT11 sensor to monitor temperature, an MQ-2 sensor to detect LPG and smoke, and a flame sensor to detect open fire. Based on the sensor data, AI-driven logic decides whether to switch ON the cooling/exhaust fan, sound the buzzer, or push an alert notification to the user's smartphone through the Blynk IoT platform, all without requiring constant human supervision.

1.2 Objectives

1. To design and develop an IoT-enabled smart kitchen safety system using the ESP32 microcontroller.
2. To continuously monitor kitchen temperature, gas concentration, smoke, and flame using DHT11, MQ-2, and flame sensors.
3. To apply AI-based decision logic that analyses sensor data to detect hazardous conditions and predict fire risk.
4. To automatically control a 12V DC fan for cooling and ventilation based on temperature and occupancy.
5. To trigger a buzzer alarm and send real-time IoT alerts to the user's mobile application during emergencies.
6. To enable remote, real-time monitoring and manual control of the system through the Blynk mobile application.
7. To reduce energy consumption through AI-based occupancy detection that switches the fan OFF when the kitchen is unoccupied.

1.3 Scope of the Project

The AI-Enhanced IoT Enabled Smart Kitchen Safety and Temperature Controlled Fan Automation System aims to provide an affordable, scalable, and intelligent safety solution suitable for residential kitchens, hostels, canteens, and small commercial kitchens. The system uses an ESP32 microcontroller integrated with a DHT11 temperature sensor, an MQ-2 gas sensor, a flame sensor, and an optional PIR sensor for occupancy detection.

Based on the collected sensor data, the system automatically controls a 12V DC fan through a relay driver stage, activates a buzzer alarm, and pushes real-time alerts to a connected mobile application. The scope of the current implementation focuses on temperature-controlled fan automation, gas/smoke/fire hazard detection, and IoT-based remote monitoring; the design can be extended in future work to include automatic gas-valve shutoff, camera-based AI vision, and integration with home-automation ecosystems.

Chapter 2 – Literature Survey

The increasing demand for home safety and smart automation has led to significant research in IoT-based kitchen and fire-safety monitoring systems. Conventional kitchen safety measures rely on passive devices such as standalone smoke detectors and manual LPG regulators, which do not provide real-time alerts or remote monitoring, resulting in delayed responses during emergencies.

Several studies have demonstrated the use of ESP32 and Arduino-based microcontrollers integrated with gas sensors such as the MQ-2 and MQ-6 for LPG and smoke detection in domestic environments. These systems typically combine gas sensors with buzzers and relay-controlled ventilation fans to reduce the concentration of hazardous gases. Researchers have also explored the use of DHT11 and DHT22 sensors for temperature and humidity monitoring, enabling automatic activation of cooling or exhaust systems when a threshold temperature is exceeded.

Recent advancements have explored the integration of Artificial Intelligence and rule-based decision engines that combine multiple sensor

inputs – temperature, gas concentration, smoke, and flame – to more accurately predict fire risk and reduce false alarms compared to single-sensor systems. IoT platforms such as Blynk, Firebase, and ThingSpeak are commonly used to enable remote monitoring, historical data logging, and instant mobile notifications, allowing users to respond to hazards even when they are away from the kitchen.

Occupancy-aware automation, using PIR motion sensors or camera-based human detection, has also been studied as a means of improving the energy efficiency of automated appliances by ensuring that fans and other actuators operate only when a person is present. Based on the findings of previous research, the AI-Enhanced IoT Enabled Smart Kitchen Safety and Temperature Controlled Fan Automation System combines AI-based multi-sensor fusion, ESP32 processing, relay-driven fan automation, and Blynk-based IoT connectivity to overcome the limitations of single-sensor, non-connected kitchen safety devices, providing a scalable and energy-efficient solution suitable for modern smart homes.

Chapter 3 – System Design / Methodology

3.1 System Architecture

The AI-Enhanced IoT Enabled Smart Kitchen Safety and Temperature Controlled Fan Automation System is built around the ESP32 microcontroller, which acts as the central processing unit for the entire architecture. Two 12V Li-ion batteries provide the primary power source, which is regulated to 5V for the ESP32 and sensor modules while directly powering the relay-driven fan circuit at 12V.

A DHT11 sensor continuously measures the ambient kitchen temperature and transmits the readings to the ESP32. The ESP32 applies AI-based threshold and rule logic to this data, along with inputs from the MQ-2 gas sensor and flame sensor, to determine whether the environment is NORMAL, WARNING, or DANGER. Based on this classification, the ESP32 drives a two-stage transistor circuit (BC547 and C1815) that switches a 1-channel relay module, which in turn controls the 12V DC fan.

Simultaneously, the ESP32 updates a 16x2 I2C LCD with the live temperature and system status, drives a buzzer during emergencies, and transmits real-time data to the Blynk IoT platform over Wi-Fi. Through the Blynk mobile application, the user can view live temperature readings, fan

status, and receive push notifications for hazardous conditions, as well as manually override the fan from anywhere with an internet connection.

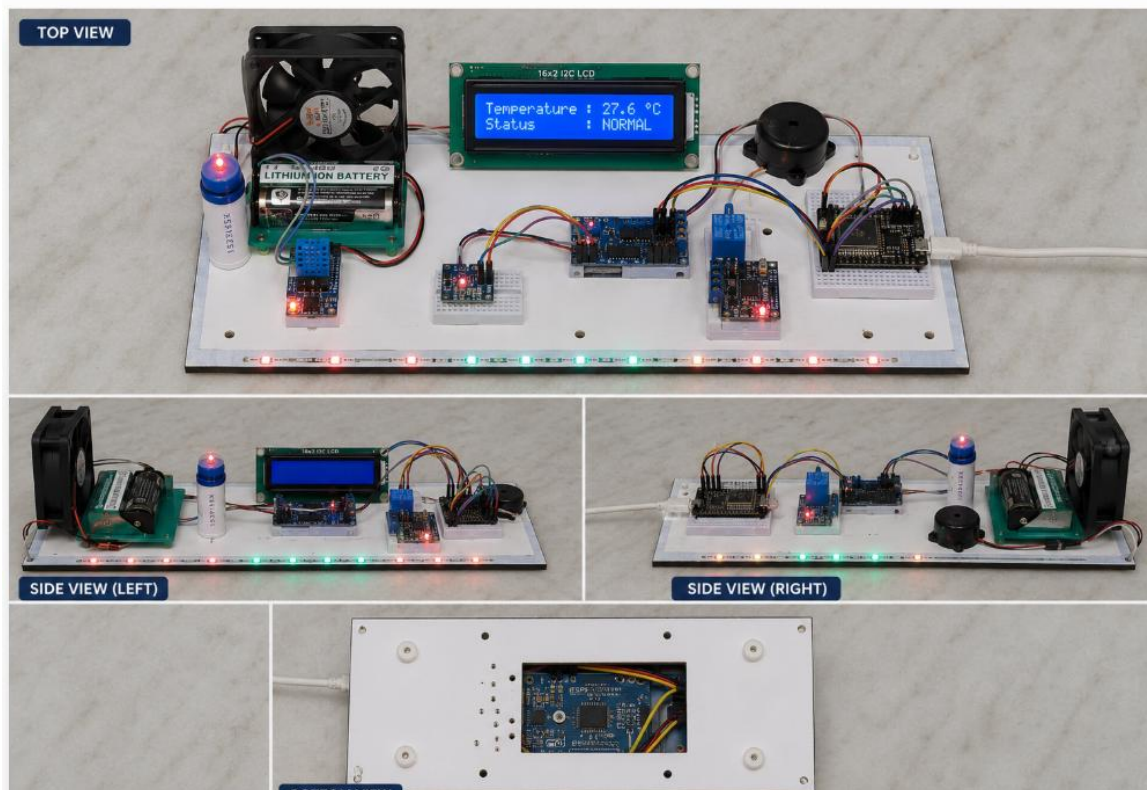


Fig. 3.1 Assembled Hardware Prototype – Top, Side, and Bottom Views

3.2 Block Diagram

The block diagram below illustrates the overall system architecture, showing the flow of data and control from the power supply and sensors through the ESP32 microcontroller, driver stage, and relay module to the output fan, along with the IoT monitoring path through the Blynk application.

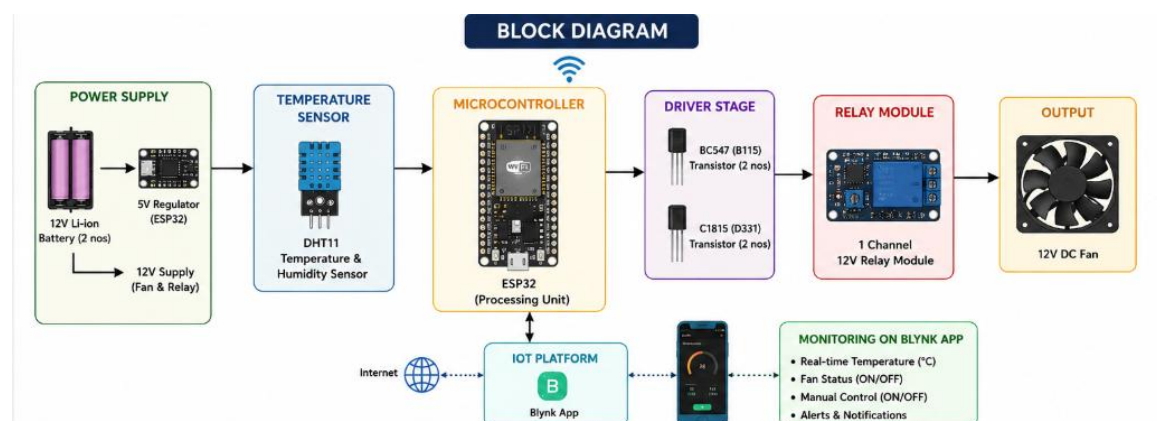


Fig. 3.2 Block Diagram of the Smart Kitchen Safety System

Working of the Block Diagram:

- **Power Supply:** Two 12V Li-ion batteries supply 12V directly to the fan and relay circuit, and 5V (via regulator) to the ESP32 and sensors.
- **Temperature Sensing:** The DHT11 sensor continuously measures kitchen temperature and humidity and sends the data to the ESP32.
- **Processing Unit:** The ESP32 applies AI-based decision logic on temperature, gas, smoke, and flame data to classify the kitchen condition.
- **Driver Stage:** BC547 and C1815 transistors amplify the ESP32's low-current GPIO signal to safely switch the relay coil.
- **Relay Module:** The 1-channel 12V relay switches power to the DC fan based on the ESP32's control signal.
- **Output:** The 12V DC fan provides cooling and ventilation when activated.
- **IoT Platform:** Sensor data, fan status, and alerts are transmitted to the Blynk App for real-time monitoring, manual control, and notifications.

3.3 Flowchart

The flow of operation begins when the system powers ON and continuously loops through sensor reading, AI-based hazard evaluation, and actuator control, as summarised below:

8. **START** – Power ON the system and initialise the ESP32, sensors, LCD, and Blynk connection.
9. Read the DHT11 temperature sensor, MQ-2 gas sensor, and flame sensor.
10. **AI Evaluation** – Check if flame is detected or gas/smoke level exceeds the safety threshold.
11. If **YES** – activate the buzzer, turn the fan ON, and send an IoT fire/gas alert notification.
12. If **NO** – check if temperature exceeds the set threshold and occupancy is detected.

13. If temperature is high and the kitchen is occupied – turn the fan ON for cooling.
14. If temperature is normal or the kitchen is unoccupied – keep the fan OFF to save energy.
15. Update the LCD display and send the latest status to the Blynk App.
16. Repeat the loop continuously – END (on power OFF).

3.4 Hardware Requirements

The hardware components used in the project, along with their specifications, quantities, and purpose, are listed below.

S.No.	Component	Specification / Qty	Purpose
1	ESP32 Dev Board	ESP32 (Wi-Fi + Bluetooth), 1 No.	Main microcontroller and processing unit
2	DHT11 Sensor	Temperature & Humidity, 1 No.	Measures kitchen temperature for AI analysis
3	MQ-2 Gas Sensor	LPG / Methane / Smoke, 1 No.	Detects gas leakage and smoke concentration
4	Flame Sensor	IR Flame Detection, 1 No.	Detects presence of open flame / fire
5	1-Channel Relay Module	12V, 1 No.	Switches the 12V DC fan ON/OFF
6	BC547 (B115) Transistor	NPN, 2 Nos.	Driver stage to switch the relay coil
7	C1815 (D331) Transistor	NPN, 2 Nos.	Driver stage to switch the relay coil

S.No.	Component	Specification / Qty	Purpose
8	1N4007 Diode	Flyback Diode, 2 Nos.	Protects transistors from relay back-EMF
9	16x2 I2C LCD	16x2 Display with I2C Module, 1 No.	Displays temperature, status and alerts
10	Buzzer	5V Piezo Buzzer, 1 No.	Sounds alarm during gas/smoke/fire emergency
11	LED Indicators	5mm LED (Red/Green/Yellow), 3 Nos.	Shows normal, warning and danger conditions
12	12V Li-ion Battery	12V Li-ion Cell, 2 Nos.	Primary power source for the system
13	Battery Holder	2-Cell Holder, 1 No.	Houses and secures the Li-ion battery pack
14	5V Regulator Module	AMS1117 / LM7805, 1 No.	Regulates 12V battery output to 5V for ESP32
15	1000µF / 50V Capacitor	Electrolytic Capacitor, 2 Nos.	Filters and stabilises the power supply
16	12V DC Fan	Brushless DC Cooling Fan, 1 No.	Provides cooling / exhaust ventilation
17	Breadboard	Standard Prototyping Board, 1 No.	Used for circuit assembly and testing
18	Male-to-Male Jumper Wires	Connecting Wires, 10 Nos.	Interconnects all modules on the breadboard

S.No.	Component	Specification / Qty	Purpose
19	Power Supply	5V / 12V Adapter	Powers the complete hardware setup

Description of Key Components

17. ESP32 Dev Board: Acts as the central controller, running the AI decision logic and managing Wi-Fi based IoT communication.
18. DHT11 Sensor: Provides real-time temperature data used to trigger the cooling fan and detect abnormal heat conditions.
19. MQ-2 Gas Sensor: Detects LPG, methane, and smoke concentration in the kitchen air for early hazard detection.
20. Flame Sensor: Detects infrared radiation from an open flame to provide instant fire detection.
21. BC547 & C1815 Transistors: Form a two-stage driver circuit that safely switches the relay coil using the ESP32's GPIO signal.
22. 1N4007 Diodes: Act as flyback diodes to protect the transistors from the relay coil's back-EMF.
23. 1-Channel Relay Module: Switches the 12V supply to the DC fan based on the ESP32's control output.
24. 16x2 I2C LCD: Displays live temperature and system status (NORMAL / WARNING / DANGER) for local monitoring.
25. Buzzer & LED Indicators: Provide audible and visual alerts corresponding to normal, warning, and danger conditions.
26. 12V Li-ion Batteries & 1000 μ F Capacitors: Provide a stable, filtered power supply for reliable operation of the fan and relay circuit.

3.5 Software Requirements

The software tools required for developing and deploying the project are listed below.

S.No.	Software	Description
1	Arduino IDE	Used to write, compile, and upload the embedded program to the ESP32 microcontroller.
2	ESP32 Board Package	Provides ESP32 board support and libraries within the Arduino IDE.
3	Embedded C/C++	Programming language used to develop sensor reading, AI decision logic, and control functions.
4	Blynk IoT Platform	Enables real-time remote monitoring, alerts, and manual control through a mobile application.
5	Serial Monitor	Displays live sensor readings and system status for debugging and testing.

3.6 AI-Enhanced Smart Features

In addition to basic threshold-based automation, the system incorporates the following AI-driven smart features to improve safety and energy efficiency:

- **AI Occupancy Detection:** Uses a PIR sensor (or camera) to detect human presence in the kitchen. The fan automatically turns OFF when the room is empty and turns ON when a person is detected, improving energy efficiency.
- **AI Gas Leak Detection:** The AI logic continuously analyses MQ-2 gas sensor data to detect LPG leaks and automatically activates the buzzer, turns ON the exhaust fan, and sends an IoT alert to the user's phone.
- **AI Smoke Detection:** The AI monitors smoke levels from the gas sensor to identify abnormal smoke conditions and triggers the fan, alarm, and a mobile notification before the situation escalates.

- **AI Fire Risk Detection:** The AI combines temperature, gas, and smoke sensor data together (sensor fusion) to predict potential fire hazards and provide an early warning through the buzzer and IoT notifications, reducing false alarms compared to a single-sensor approach.

Together, these AI-enhanced features transform the system from a simple threshold-based controller into an intelligent kitchen safety assistant capable of anticipating hazards, reducing false alarms, and optimising energy usage based on real occupancy.

3.7 Development Process

The development of the project was carried out in the following stages:

Step 1: Analyse the project requirements and identify safety and automation objectives.

Step 2: Select the required hardware components – ESP32, DHT11, MQ-2, flame sensor, relay module, transistors, fan, and battery pack.

Step 3: Design and assemble the circuit, including the power supply, sensor interfacing, and transistor-driven relay stage.

Step 4: Develop the embedded program in Arduino IDE, implementing sensor reading and AI-based decision logic.

Step 5: Integrate the Blynk IoT platform for real-time monitoring, alerts, and manual control.

Step 6: Test the system under normal, high-temperature, gas-leak, smoke, and flame conditions.

Step 7: Verify system performance, calibrate thresholds, and optimise the design for reliability and energy efficiency.

Chapter 4 – Implementation

4.1 Project Development

The AI-Enhanced IoT Enabled Smart Kitchen Safety and Temperature Controlled Fan Automation System was developed through a structured process to achieve an intelligent, safe, and energy-efficient kitchen automation solution. Development began with identifying the project objectives and selecting suitable hardware components, including the

ESP32 microcontroller, DHT11 sensor, MQ-2 gas sensor, flame sensor, relay module, driver transistors, and the 12V DC fan.

A circuit was designed and assembled on a breadboard/PCB to establish communication between the sensors, the ESP32, and the output actuators (fan, buzzer, LCD). The system software was then developed using the Arduino IDE, where the ESP32 was programmed to read sensor data, apply AI-based rule logic, and control the fan, buzzer, and LCD display automatically. IoT functionality was integrated using the Blynk platform to enable remote monitoring, manual override, and real-time alert notifications.

After hardware and software integration, the complete system was assembled onto a compact wooden/acrylic base plate housing the battery pack, sensor modules, ESP32, relay, buzzer, and LCD, as shown in the hardware prototype photographs. The system was then tested under a range of environmental conditions to verify its performance, energy efficiency, and reliability, and necessary calibration was performed based on the test results.

4.2 Circuit Diagram

The circuit consists of the following connections and stages:

27. Power Supply: Two 12V Li-ion batteries, filtered by 1000 μ F/50V capacitors, supply 12V directly to the relay and fan, and 5V (via a regulator module) to the ESP32 and DHT11 sensor.
28. ESP32 Connections: The ESP32 reads data from the DHT11 sensor and drives two GPIO outputs (GPIO 25 and GPIO 26) into the transistor driver stage.
29. Driver Stage: BC547 (B115) and C1815 (D331) transistors, each with a 1k Ω base resistor, switch the relay coil based on the ESP32's control signal; 1N4007 diodes protect the transistors from back-EMF.
30. Relay Module: The 1-channel 12V relay module switches the 12V supply to the DC fan when triggered.
31. Output: The 12V DC fan turns ON for cooling/ventilation when the relay is activated, based on temperature, gas, or flame conditions.

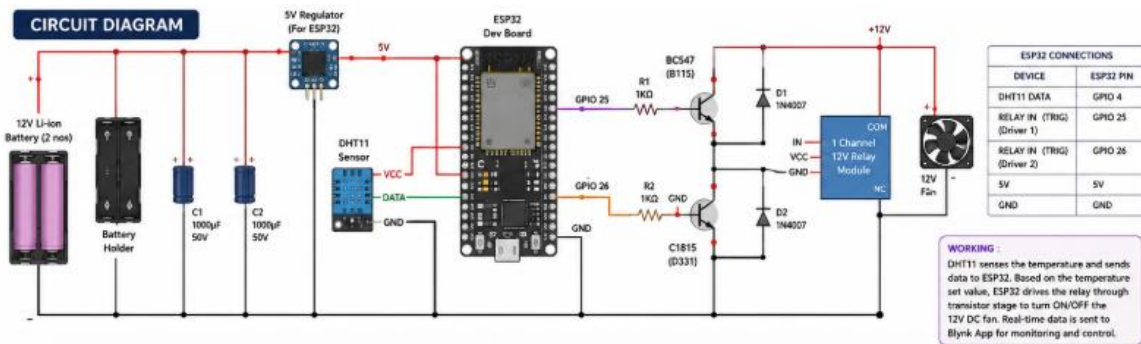


Fig. 4.1 Complete Circuit Diagram

ESP32 Pin Connections

Device	ESP32 Pin
DHT11 Data	GPIO 4
Relay IN (Trigger 1 / Driver 1)	GPIO 25
Relay IN (Trigger 2 / Driver 2)	GPIO 26
Relay Module VCC	5V
Relay Module GND	GND

Circuit Description

The AI-Enhanced Smart Kitchen Safety circuit is designed around the ESP32 microcontroller, which serves as the central control unit. A regulated 5V supply (derived from the 12V Li-ion battery pack via a 5V regulator) powers the ESP32 and DHT11 sensor, while the batteries directly supply 12V to the relay and fan circuit. The DHT11 sensor continuously reports temperature to the ESP32 over GPIO 4. Based on the AI-based decision logic – which also factors in gas, smoke, and flame readings – the ESP32 drives GPIO 25 and GPIO 26 high to switch the BC547 and C1815 transistors ON, energising the relay coil and turning the 12V DC fan ON. The ESP32 also utilises its built-in Wi-Fi capability to transmit live data to the Blynk IoT platform for remote monitoring, alerts, and manual override, making the overall circuit efficient, reliable, and suitable for smart kitchen safety applications.

4.3 ESP32 Program (Source Code)

Below is a simplified Arduino IDE (ESP32) program suitable for inclusion in the project report. It demonstrates the working of the AI-Enhanced IoT Enabled Smart Kitchen Safety and Temperature Controlled Fan Automation System using the DHT11, MQ-2, flame, and PIR sensors, with fan, buzzer, and Blynk IoT integration.

```
// AI-Enhanced IoT Enabled Smart Kitchen Safety and
```

```
// Temperature Controlled Fan Automation System
```

```
// ESP32 Arduino Code (Simplified for Report)
```

```
#include <DHT.h>
```

```
#include <BlynkSimpleEsp32.h>
```

```
#define DHT_PIN 4
```

```
#define DHT_TYPE DHT11
```

```
#define GAS_PIN 34    // MQ-2 Gas Sensor (Analog)
```

```
#define FLAME_PIN 27  // Flame Sensor (Digital)
```

```
#define PIR_PIN 14    // PIR Occupancy Sensor
```

```
#define RELAY_PIN1 25 // Fan Relay Driver 1
```

```
#define RELAY_PIN2 26 // Fan Relay Driver 2
```

```
#define BUZZER_PIN 33
```

```
DHT dht(DHT_PIN, DHT_TYPE);
```

```
float tempThreshold = 35.0; // AI temperature threshold (°C)
```

```
int gasThreshold = 1800; // AI gas / smoke threshold
```

```
void setup() {  
  Serial.begin(115200);  
  dht.begin();  
  pinMode(FLAME_PIN, INPUT);  
  pinMode(PIR_PIN, INPUT);  
  pinMode(RELAY_PIN1, OUTPUT);  
  pinMode(RELAY_PIN2, OUTPUT);  
  pinMode(BUZZER_PIN, OUTPUT);  
  Blynk.begin(auth, ssid, pass);  
}
```

```
void loop() {  
  Blynk.run();  
  
  float temperature = dht.readTemperature();  
  int gasLevel = analogRead(GAS_PIN);  
  int flame = digitalRead(FLAME_PIN);  
  int occupancy = digitalRead(PIR_PIN);  
  
  // AI Occupancy Detection  
  bool personPresent = (occupancy == HIGH);  
  
  // AI Fire Risk Score (combines temp + gas + flame)  
  bool fireRisk = (temperature > tempThreshold && gasLevel > gasThreshold)  
    || (flame == LOW);
```

```
if (fireRisk) {
  digitalWrite(BUZZER_PIN, HIGH);
  digitalWrite(RELAY_PIN1, HIGH); // Fan ON
  digitalWrite(RELAY_PIN2, HIGH);
  Blynk.logEvent("fire_alert", "Fire risk detected in kitchen!");
}

else if (gasLevel > gasThreshold) {
  digitalWrite(BUZZER_PIN, HIGH); // AI Gas Leak Detection
  digitalWrite(RELAY_PIN1, HIGH);
  digitalWrite(RELAY_PIN2, HIGH);
  Blynk.logEvent("gas_alert", "Gas leak detected!");
}

else if (temperature > tempThreshold && personPresent) {
  digitalWrite(RELAY_PIN1, HIGH); // Fan ON for cooling
  digitalWrite(RELAY_PIN2, HIGH);
  digitalWrite(BUZZER_PIN, LOW);
}

else {
  digitalWrite(RELAY_PIN1, LOW); // AI Occupancy: fan OFF if empty
  digitalWrite(RELAY_PIN2, LOW);
  digitalWrite(BUZZER_PIN, LOW);
}

Blynk.virtualWrite(V0, temperature);
Blynk.virtualWrite(V1, gasLevel);
Serial.print("Temp: "); Serial.println(temperature);
```

```
delay(1000);  
}
```

4.4 Working Principle

The AI-Enhanced IoT Enabled Smart Kitchen Safety and Temperature Controlled Fan Automation System operates by continuously monitoring kitchen conditions and automatically responding to unsafe or uncomfortable situations. The ESP32 microcontroller periodically reads the DHT11 temperature sensor, the MQ-2 gas sensor, the flame sensor, and (optionally) a PIR occupancy sensor.

The AI-based logic first checks for high-priority hazards – an open flame or a gas/smoke level above the safety threshold. If either is detected, the buzzer is activated immediately, the fan is switched ON to help disperse gas or smoke, and an IoT alert notification is pushed to the user's mobile application through Blynk. If no immediate hazard is present, the system checks the kitchen temperature against a configurable threshold together with the occupancy status; the fan is switched ON only when the temperature is high and the kitchen is occupied, and switched OFF when the kitchen is empty, thereby conserving energy.

Throughout operation, the current temperature and system status (NORMAL, WARNING, or DANGER) are displayed on the 16x2 I2C LCD for local monitoring, while the same data, along with fan status and alerts, is transmitted to the Blynk App for real-time remote monitoring and manual control. This combination of multi-sensor AI logic and IoT connectivity allows the system to react faster and more reliably than a manual or single-sensor approach.

4.5 Implementation Steps

Step 1: Collect the required hardware and software components.

Step 2: Assemble the circuit using the ESP32, DHT11, MQ-2, flame sensor, transistor driver stage, relay, and 12V DC fan.

Step 3: Develop and upload the AI-based control program to the ESP32 using the Arduino IDE.

Step 4: Test the system for temperature-based fan control, gas/smoke detection, and flame detection.

Step 5: Integrate the Blynk IoT platform and verify real-time monitoring, alerts, and manual override.

Step 6: Mount all components onto the base plate and verify the complete hardware assembly, as shown in the prototype photographs.

4.6 Hardware Prototype

The completed hardware prototype, including the top view (showing the battery, fan, DHT11 sensor, LCD, buzzer, ESP32, and relay module), the side views, and the bottom view (showing the ESP32 mounted on the underside of the base plate), is presented in Fig. 3.1. The compact layout ensures that all sensors are exposed to the kitchen environment while the processing and driver electronics remain protected.

Chapter 5 – Results and Discussion

5.1 Output

The AI-Enhanced IoT Enabled Smart Kitchen Safety and Temperature Controlled Fan Automation System successfully monitors kitchen temperature, gas concentration, smoke, and flame in real time. During testing, the system correctly displayed the live temperature and status (e.g., “Temperature: 27.6°C, Status: NORMAL”) on the 16x2 I2C LCD, as shown in the hardware prototype. The fan was automatically switched ON when the temperature exceeded the configured threshold or when a gas/smoke/flame hazard was detected, and switched OFF automatically when conditions returned to normal or the kitchen became unoccupied.

The buzzer and LED indicators correctly reflected the NORMAL, WARNING, and DANGER states, and all sensor readings, fan status, and alerts were successfully transmitted to and displayed on the Blynk mobile application, confirming reliable IoT-based remote monitoring and control.

5.2 Testing

The project was tested under several operating conditions to evaluate its performance, as summarised in the table below.

Test Case	Expected Result	Actual Result	Status
Power ON	System initialises and LCD displays status	System initialised, LCD displayed "NORMAL"	Pass
Normal Temperature	Fan remains OFF, status NORMAL	Fan stayed OFF, status displayed NORMAL	Pass
High Temperature	Fan turns ON automatically	Fan turned ON at threshold temperature	Pass
Gas Leak Simulation	Buzzer ON, fan ON, IoT alert sent	Buzzer activated, fan turned ON, alert received on app	Pass
Smoke Detection	Alarm and notification triggered	Alarm triggered, notification received	Pass
Flame Detection	Immediate alert and buzzer activation	Buzzer activated instantly, alert sent	Pass
Occupancy Detection	Fan OFF when kitchen is empty	Fan switched OFF after occupant left	Pass
Remote Monitoring	Live data visible on Blynk App	Temperature and status updated in real time	Pass

The testing confirmed that the system operated reliably and according to the design requirements across all evaluated scenarios.

5.3 Observations

- 32. The system accurately measured and displayed the kitchen temperature on the LCD in real time.
- 33. The fan was automatically activated whenever the temperature exceeded the configured AI threshold.
- 34. The MQ-2 sensor reliably detected simulated LPG/smoke conditions, triggering the buzzer and fan.
- 35. The flame sensor provided an instantaneous response to a simulated open flame, activating the alarm within a fraction of a second.
- 36. AI-based occupancy logic successfully switched the fan OFF when no motion was detected, reducing unnecessary power consumption.
- 37. All sensor data, fan status, and alerts were successfully synchronised with the Blynk mobile application with minimal latency.
- 38. The overall system demonstrated stable and energy-efficient operation throughout the testing period.

5.4 Advantages

- 39. Provides continuous, automated monitoring of kitchen temperature, gas, smoke, and flame conditions.
- 40. Reduces fire and gas-leak risk through fast, AI-based multi-sensor hazard detection.
- 41. Improves energy efficiency through AI-based occupancy detection that switches the fan OFF when the kitchen is empty.
- 42. Enables real-time remote monitoring, alerts, and manual control via the Blynk IoT mobile application.
- 43. Reduces dependence on constant manual supervision, improving safety for elderly or distracted occupants.

44. Offers a low-cost, compact, and scalable design suitable for homes, hostels, and small commercial kitchens.

Chapter 6 – Conclusion and Future Scope

6.1 Conclusion

The AI-Enhanced IoT Enabled Smart Kitchen Safety and Temperature Controlled Fan Automation System successfully demonstrates an intelligent, safe, and energy-efficient kitchen automation solution using the ESP32 microcontroller, DHT11, MQ-2, and flame sensors, together with AI-based decision logic and IoT connectivity through the Blynk platform. The system automatically controls the cooling/exhaust fan based on temperature and occupancy, while instantly responding to gas leakage, smoke, and fire hazards with buzzer alarms and mobile notifications.

The project offers a cost-effective, reliable, and scalable solution for residential and small commercial kitchens, and highlights the practical application of AI-driven sensor fusion and IoT technology in improving everyday domestic safety and reducing energy consumption.

6.2 Future Scope

- Integration of an automatic LPG solenoid valve to physically shut off the gas supply when a leak is detected.
- Use of a camera module with an on-device AI vision model for more accurate occupancy and hazard detection (e.g., detecting unattended cooking).
- Cloud-based data logging and analytics to identify usage patterns and predict maintenance needs.
- Integration with home automation ecosystems (Google Home / Alexa) for voice-based control and status queries.
- Addition of a battery-backed GSM module to send SMS alerts during Wi-Fi or internet outages.

6.3 References

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6. IR Flame Sensor Module Datasheet.
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